

Open-World Novelty on Transfer Datasets: End-to-End Fine-Tuning, Discovery, and Residual Training

SuperSig experiments 70–74 (with context from 49/51/62/69)

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Abstract

We evaluate six contrastive/regularized pretraining objectives — three unsupervised (SimCLR, SIGReg-SSL/LeJEPa, unsupervised NPLM) and three supervised (SupCon, SupCon+SIGReg, supervised NPLM) — as *end-to-end fine-tuning* losses for ViT-B/16 trunks on four transfer datasets (Stanford Cars, Flowers-102, DTD, Galaxy10 DECals), each from three pretrained bases (DINO, LeJEPa, VISReg): a $4 \times 3 \times 6$ grid. Every space is scored with an open-world novelty battery (held-out-class linear probe, nearest-centroid accuracy, distance-based novelty AUCs, and dataset-level anomaly-detection power), before and after an iterated anchor-*discovery* phase, and finally under *residual* fine-tuning. Three headline results: (i) discovery, which is probe-negative on frozen trunks, becomes probe-positive in 50/72 grid cells once the trunk is fine-tuned end-to-end — and the SupCon-ft+discovery pipeline is positive in 12/12 dataset×base cells; (ii) the champion pre-discovery objective is regime-determined and base-invariant (SupCon+SIGReg on fine-grained data, SimCLR on textures, SupCon on few-class coarse data, supervised NPLM for calibration at few classes); (iii) residual fine-tuning beats the discovery pipeline in all 12 cells, with the NPLM residual concat uniquely raising the probe *while retaining* calibrated per-event power — the first cars space to break the probe-calibration dissociation; (iv) a final discovery pass on the residual winners is *purity-gated*: it stacks only where the residual space separates the novelty, yielding the program’s final records — aircraft 0.863, cars 0.855, flowers 0.906, DTD 0.862, Galaxy10 0.975 — every one a residual pipeline; (v) the same recipe transfers back to CIFAR (exps 73–74): supcon→res concat sets a new CIFAR-100 probe record (0.9594) and ties the CIFAR-10 pre-discovery record (0.9536), and discovery on these concats is again a calibration repair at low purity, not a probe booster.

1 Setup

Datasets and open-world protocol. Each dataset’s final classes are held out entirely: their images and labels are removed from every fine-tuning corpus, so held-out classes are genuinely novel at training time.

dataset	classes	held out	train pool (seen)	character
Stanford Cars	196	186–195	7 712	fine-grained (car models)
Flowers-102	102	92–101	1 840	fine-grained, few-shot (~ 18 /class)
DTD	47	37–46	2 960	texture
Galaxy10 DECals	10	{9}	1 585	coarse morphology, 10%/90% split

Table 1: Datasets. Bases: DINO, LeJEPa and VISReg ViT-B/16 trunks (768-D CLS features).

Fine-tuning recipe (exp-49/62 convention). Model = pretrained trunk + projection head $768 \rightarrow 256 \rightarrow 100$; everything trainable. Adam with cosine schedule, trunk lr 10^{-5} , head lr 10^{-3} , AMP fp16, 20 epochs, batch 32 with two augmented views (RandomResizedCrop, flip, jitter at 224px). Per-arm seeds. All objectives operate on the raw 100-D head output $\mathbf{z}$; only softmax-contrastive losses ℓ_2 -normalize internally.

2 Loss functions

Let $\mathbf{z}_i = f_\theta(\mathbf{x}_i) \in \mathbb{R}^{100}$, with two views per image stacked into a $2N$ batch; $\hat{\mathbf{z}} = \mathbf{z}/\|\mathbf{z}\|$.

2.1 Core primitives

SIGReg (sliced Gaussianity). The repo’s member of the VISReg/LeJEPa family: a sliced 1-D Wasserstein-2 distance to $\mathcal{N}(0, I)$,

$$\mathcal{L}_{\text{SIG}}(\mathbf{z}_{1:B}) = \frac{1}{S} \sum_{s=1}^S \frac{1}{B} \sum_{b=1}^B (\text{sort}_b(\langle \mathbf{z}, u_s \rangle) - \Phi^{-1}(q_b))^2, \quad u_s \sim \text{Unif}(\mathbb{S}^{d-1}), \quad (1)$$

with $S=64$ slices and $\Phi^{-1}(q_b)$ the standard-normal quantiles. A *classwise* variant applies the same functional per class to $\mathbf{z} - \boldsymbol{\mu}_c$, pulling class c toward $\mathcal{N}(\boldsymbol{\mu}_c, I)$ (used in the CIFAR program; not part of the transfer grid below).

Softmax contrastive (NT-Xent / SupCon). For positives $P(i)$ and temperature τ ,

$$\mathcal{L}_{\text{con}} = \sum_i \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\langle \hat{\mathbf{z}}_i, \hat{\mathbf{z}}_p \rangle / \tau)}{\sum_{a \neq i} \exp(\langle \hat{\mathbf{z}}_i, \hat{\mathbf{z}}_a \rangle / \tau)}. \quad (2)$$

With instance positives (the other view) and $\tau=0.5$ this is SimCLR/NT-Xent; with class positives and $\tau=0.1$ it is SupCon.

2.2 The hybrid contrastive cube (NPLM losses)

HybridContrastiveLoss factorizes a contrastive objective into *positives* $\times$ *critic* $\times$ *estimator* $\times$ *marginal*, $\mathcal{L} = \mathcal{L}_{\text{int}}(g) + \lambda \mathcal{L}_{\text{marg}}(\mathbf{z})$:

- **critic** $g(\mathbf{z}, \mathbf{z}')$: cosine $\langle \hat{\mathbf{z}}, \hat{\mathbf{z}}' \rangle / \tau$; bilinear $\langle \mathbf{z}, \mathbf{z}' \rangle / \tau$; distance $-\frac{1}{2\tau} \|\mathbf{z} - \mathbf{z}'\|^2$.
- **estimator**: *softmax* (self-normalized per row, as above — recovers pointwise mutual information only up to a per-row constant), or *NPLM*: the un-normalized maximum-likelihood objective

$$\mathcal{L}_{\text{NPLM}} = \mathbb{E}_{\text{ref}}[e^g - 1] - \mathbb{E}_{\text{pos}}[g], \quad (3)$$

whose minimizer is the *absolute* log density ratio $g^* = \log \frac{p(\mathbf{x}, \mathbf{x}')}{p(\mathbf{x})p(\mathbf{x}')}$ with the calibration property $\mathbb{E}_{\text{ref}}[e^{g^*}] = 1$ (verified empirically, $r > 0.9$ against analytic PMI). The reference set excludes positives and the diagonal; the exponent is clamped at 30 (row-max subtraction would destroy calibration). At initialization the interaction term is $\mathcal{O}(10^{10})$, so λ is inert early and only shapes the endgame; $\tau = 1$ is a sharp optimum for the bilinear critic.

- **marginal**: none, global SIGReg, or classwise SIGReg.

Calibrated NPLM spaces carry *absolute* distance information (a ready-to-use novelty metric), where softmax spaces are the better linear probe inputs — the program’s central probe-calibration dissociation.

2.3 The six fine-tuning arms (exp 70)

arm	objective	labels	knobs
simclr-ft	NT-Xent	no	$\tau=0.5$
sigreg-ssl-ft	$\ \mathbf{z}_1 - \mathbf{z}_2\ ^2$ (MSE inv.) + SIGReg	no	LeJEPa objective, $\lambda=1$
nplm-bil-ft	NPLM, instance/bilinear + SIGReg	no	$\tau=1, \lambda=1$
supcon-ft	SupCon	yes	$\tau=0.1$
ss-ft	SupCon + SIGReg	yes	$\tau=0.1, \lambda=5$
nplm-sup-ft	NPLM, supervised/distance + SIGReg	yes	$\tau=1, \lambda=1$

Table 2: Pretraining arms: {SimCLR, SIGReg, NPLM} families $\times$ {unsupervised, supervised}.

2.4 Residual objectives (exp 71)

Given a trained supervised *parent* with frozen seen-class centroids $\boldsymbol{\mu}_y$ in head space, a deepcopy is fine-tuned end-to-end on the residual $\mathbf{r} = \mathbf{z} - \boldsymbol{\mu}_y$:

$$\text{res:} \quad \mathcal{L} = \text{NTXent}_{\tau=0.5}(\hat{\mathbf{r}}) + 5 \mathcal{L}_{\text{SIG}}(\mathbf{r}) \quad (\text{exp-36/49 champion residual objective}), \quad (4)$$

$$\text{res-nplm:} \quad \mathcal{L} = \mathcal{L}_{\text{NPLM}}^{\text{inst/bilinear}}(\mathbf{r}) + \mathcal{L}_{\text{SIG}}(\mathbf{r}) \quad (\text{exp-59 NPLM residual corner}). \quad (5)$$

Constructions per cell: supcon-ft $\rightarrow$ res, ss-ft $\rightarrow$ res, supcon-ft $\rightarrow$ res-nplm; each evaluated as the residual head space alone and as the 200-D concat [parent; residual].

3 Training strategies and evaluation

End-to-end fine-tuning (exp 70). Each arm fine-tunes trunk+head on the *seen-class* two-view corpus (held-out classes excluded — images and labels), unlike earlier closed-set fine-tunes, so all novelty numbers are strictly open-world.

Discovery (iterated anchor discovery). With the fine-tuned trunk frozen as a feature bank and the head as the discovery backbone: (1) embed the train pool and pool events beyond the 0.95 distance quantile to the current anchors; (2) cluster the pool with BIC-selected k -means, creating new anchors (merged at distance < 3); (3) pseudo-label and fine-tune the head with the proto/repulse hybrid (prototype cross-entropy on $-\frac{1}{2}d^2$ to anchors + repulsion + SIGReg), balanced batches, 2 rounds $\times$ 5 epochs. Reported both “naturally” (train pool as-is) and on an injected grid (signal fraction f swept) for post-discovery power.

Metric battery. *probe*: AUC of a linear probe trained to separate held-out from seen test images (3 seeds); *acc*: seen-class nearest-centroid accuracy; *supAUC*: seen-class supervised separability; *eucl/mahaT*: novelty AUC of min Euclidean/tied-covariance Mahalanobis distance to seen centroids; *perevt*: per-event detection power at $\alpha=0.05$ from the distance score; SparKer/Maha/MMD: dataset-level two-sample test power at injected fraction f (toy size $n_d=1000$ –2000; SparKer uses an annealed median-heuristic kernel scale).

arm (DINO base)	cars		flowers		dtd		galaxy10	
	probe	mahaT	probe	mahaT	probe	mahaT	probe	mahaT
simclr-ft	0.624	0.476	0.743	0.673	0.808	0.506	0.868	0.730
sigreg-ssl-ft	0.584	0.496	0.732	0.706	0.803	0.577	0.914	0.764
nplm-bil-ft	0.591	0.496	0.664	0.646	0.682	0.479	0.839	0.712
supcon-ft	0.733	0.564	0.708	0.873	0.799	0.646	0.938	0.786
ss-ft	0.719	0.547	0.795	0.883	0.782	0.742	0.872	0.691
nplm-sup-ft	0.549	0.547	0.625	0.668	0.586	0.615	0.878	0.824

Table 3: Pre-discovery battery, DINO base (LeJEPA/VISReg bases reproduce the same ordering). The pre-probe champion is regime-determined: fine-grained $\rightarrow$ ss-ft, texture $\rightarrow$ simclr-ft, few-class coarse $\rightarrow$ supcon-ft; supervised NPLM takes the calibration crown exactly at few classes (galaxy10: best eucl/mahaT, per-event 0.35 vs. supcon’s 0.02, round-1 pool purity 0.46–0.50 on all three bases).

4 Results

4.1 Pre-discovery: the six arms across the grid

4.2 Discovery: pre $\rightarrow$ post probe across all bases

arm	cars			flowers		
	DINO	LeJEPA	VISReg	DINO	LeJEPA	VISReg
supcon-ft	.733 $\rightarrow$.767	.699 $\rightarrow$.726	.707 $\rightarrow$.759	.708 $\rightarrow$.814	.678 $\rightarrow$.752	.714 $\rightarrow$.810
ss-ft	.719 $\rightarrow$.747	.709 $\rightarrow$.699	.741 $\rightarrow$.712	.795 $\rightarrow$.788	.792 $\rightarrow$.613	.754 $\rightarrow$.743
simclr-ft	.624 $\rightarrow$.684	.618 $\rightarrow$.614	.610 $\rightarrow$.657	.743 $\rightarrow$.754	.725 $\rightarrow$.740	.746 $\rightarrow$.728
sigreg-ssl-ft	.584 $\rightarrow$.690	.563 $\rightarrow$.617	.574 $\rightarrow$.634	.732 $\rightarrow$.777	.684 $\rightarrow$.719	.738 $\rightarrow$.734
nplm-bil-ft	.591 $\rightarrow$.704	.611 $\rightarrow$.658	.576 $\rightarrow$.624	.664 $\rightarrow$.808	.768 $\rightarrow$.723	.737 $\rightarrow$.737
nplm-sup-ft	.549 $\rightarrow$.600	.582 $\rightarrow$.658	.579 $\rightarrow$.549	.625 $\rightarrow$.624	.734 $\rightarrow$.706	.631 $\rightarrow$.576

arm	dtd			galaxy10		
	DINO	LeJEPA	VISReg	DINO	LeJEPA	VISReg
supcon-ft	.799 $\rightarrow$.811	.755 $\rightarrow$.790	.750 $\rightarrow$.791	.938 $\rightarrow$.939	.919 $\rightarrow$.931	.939 $\rightarrow$.944
ss-ft	.782 $\rightarrow$.785	.787 $\rightarrow$.784	.784 $\rightarrow$.785	.872 $\rightarrow$.926	.803 $\rightarrow$.871	.865 $\rightarrow$.945
simclr-ft	.808 $\rightarrow$.802	.817 $\rightarrow$.821	.854 $\rightarrow$.848	.868 $\rightarrow$.942	.902 $\rightarrow$.946	.933 $\rightarrow$.922
sigreg-ssl-ft	.803 $\rightarrow$.802	.809 $\rightarrow$.826	.808 $\rightarrow$.790	.914 $\rightarrow$.931	.927 $\rightarrow$.940	.931 $\rightarrow$.940
nplm-bil-ft	.682 $\rightarrow$.786	.746 $\rightarrow$.745	.778 $\rightarrow$.811	.839 $\rightarrow$.928	.893 $\rightarrow$.905	.802 $\rightarrow$.904
nplm-sup-ft	.586 $\rightarrow$.781	.581 $\rightarrow$.765	.569 $\rightarrow$.707	.878 $\rightarrow$.912	.899 $\rightarrow$.897	.883 $\rightarrow$.933

Table 4: Held-out probe, pre $\rightarrow$ post discovery, full $4 \times 3 \times 6$ grid. Discovery is positive in 50/72 cells; **supcon-ft+discovery is positive in 12/12 dataset $\times$ base cells** (the universally safe pipeline). On *frozen* trunks (exp 69, cars) discovery was probe-negative for every arm — the substrate, not the dataset, decides. Notable: the nplm-sup-ft “rescue” on DTD (+0.14 to +0.20 on every base); the one catastrophic case is ss-ft on flowers/LeJEPA (−0.18).

4.3 Residual fine-tuning (exp 71)

The flagship space. Cars/VISReg supcon $\rightarrow$ res-nplm concat: probe 0.855, nearest-centroid acc 0.608, eucl 0.747, mahaT 0.678, per-event 0.263 — the best cars value in *every column simultaneously*, and the first cars space to break the probe–calibration dissociation. In general the NPLM residual concat raises the probe while *keeping* the parent’s calibration, whereas plain-res concats often buy the probe with calibration (galaxy10/LeJEPA: probe 0.975 with per-event 0.001).

dataset	base	best construction	probe	perevt	parent	prior champ
aircraft	DINO	supcon→res-nplm concat	0.816	0.225	0.761	0.812 [†]
aircraft	LeJEPA	supcon→res-nplm concat	0.848	0.198	0.799	0.812 [†]
aircraft	VISReg	supcon→res-nplm concat	0.863	0.363	0.809	0.812 [†]
cars	DINO	supcon→res-nplm concat	0.821	0.148	0.733	0.767
cars	LeJEPA	supcon→res-nplm concat	0.779	0.082	0.699	0.726
cars	VISReg	supcon→res-nplm concat	0.855	0.263	0.707	0.759
flowers	DINO	supcon→res-nplm concat	0.885	0.334	0.708	0.814
flowers	LeJEPA	ss→res concat	0.813	0.258	0.792	0.752
flowers	VISReg	supcon→res-nplm concat	0.858	0.455	0.714	0.810
dttd	DINO	supcon→res concat	0.845	0.045	0.799	0.811
dttd	LeJEPA	ss→res concat	0.831	0.158	0.787	0.826
dttd	VISReg	supcon→res concat	0.847	0.077	0.750	0.854*
galaxy10	DINO	supcon→res concat	0.955	0.021	0.938	0.942
galaxy10	LeJEPA	supcon→res concat	0.975	0.001	0.919	0.946
galaxy10	VISReg	supcon→res residual	0.965	0.243	0.939	0.945

Table 5: Best residual construction per cell (probe; per-event power at $\alpha=0.05$), against its parent and the best prior (exp-70) pipeline. *DTD’s prior champion is the *pre*-discovery simclr-ft space — the only exp-70 record residual training does not beat. [†]Aircraft’s prior best carried a trunk-contamination caveat (the exp-62 trunks saw held-out images and labels); the aircraft rows here use new strictly open-world parents (exp-70 protocol, holdouts 90–99 excluded), so 0.863 is both a record and a stricter number. Residual fine-tuning beats the discovery pipeline in **12/12 cells** (15/15 including aircraft).

4.4 Discovery on the residual winners (exp 72)

The feature-space discovery loop was finally run on each cell’s exp-71 champion (for concat spaces the discovery backbone is a two-head wrapper — parent head on the parent-trunk bank, residual head on the child-trunk bank — and both heads fine-tune).

Aircraft sharpens the purity gate. Discovery on the three aircraft res-nplm-concat winners is probe-negative (−0.012 to −0.037) despite *moderate* pool purity (0.32–0.53), while buying geometry and per-event power (VISReg: per-event 0.30–0.50 at every fraction, mahaT +0.017): purity is necessary but *not sufficient* — on fine-grained data the res-nplm concat’s probe directions are exactly what the discovery fine-tune erodes.

Final dataset probe records (all residual pipelines). aircraft **0.863** (VISReg supcon→res-nplm concat, exp 71, strict open-world parents); cars **0.855** (VISReg supcon→res-nplm concat, exp 71); flowers **0.906** (DINO supcon→res-nplm concat + discovery, exp 72); DTD **0.862** (VISReg supcon→res concat + discovery, exp 72); Galaxy10 **0.975** (LeJEPA supcon→res concat, exp 71).

4.5 The recipe comes home: CIFAR-10/100 (exps 73–74)

The identical residual fine-tuning recipe was applied in the CIFAR world (hub-pretrained ResNet-20 backbones, holdout class 4, parents reproduced at their original exp-50 seeds; CIFAR-10 at 32+32-D, CIFAR-100 at 100+100-D):

Discovery on these concats (exp 74; image-space loop, two-net concat backbone) repeats the exp-72/74 law: pool purity is ≤ 0.03 at 64–200-D, probes stay flat (± 0.007), and the fine-tune instead repairs calibration — CIFAR-10 res-nplm concat eucl 0.712 $\rightarrow$ 0.794 with Mahalanobis-battery power 0.80 at $f=0.02$ (1.0 at $f=0.1$ for every statistic); CIFAR-100 res-nplm concat eucl

cell	space	probe pre → post	mahaT pre → post	purity r_1
flowers/DINO	supcon→res-nplm concat	0.885 → 0.906	0.843 → 0.853	0.609
flowers/VISReg	supcon→res-nplm concat	0.858 → 0.866	0.870 → 0.875	0.643
dtd/VISReg	supcon→res concat	0.847 → 0.862	0.758 → 0.778	0.292
dtd/LeJEPA	ss→res concat	0.831 → 0.838	0.791 → 0.796	0.588
cars/LeJEPA	supcon→res-nplm concat	0.779 → 0.787	0.514 → 0.534	0.104
cars/DINO, VISReg	res-nplm concats	flat (−0.001/−0.004)	~flat	0.16/0.26
flowers/LeJEPA	ss→res concat	flat	flat	0.439
dtd/DINO	supcon→res concat	0.845 → 0.838	0.604 → 0.646	0.135
galaxy10 (3 bases)	res concats / residual	flat to −0.013	mixed	0.00–0.34

Table 6: Exp 72: discovery applied to the best residual space of each cell. Gains are *purity-gated*: discovery stacks only where the residual space separates the novelty (flowers, DTD), and saturates elsewhere (cars, galaxy10 — their exp-71 records stand). New records: flowers 0.906 (first flowers space past 0.90) and DTD 0.862, retiring simclr-ft’s 0.854 — the last pre-discovery record. Calibration highlight: galaxy10/VISReg residual + discovery holds per-event power 0.32–0.45 at every injected fraction including $f=0.003$, the best low- f per-event on record.

space	CIFAR-10 (32+32)		CIFAR-100 (100+100)	
	probe	perevt	probe	perevt
supcon (parent)	0.9404	0.000	0.9577*	0.000
supcon→res concat	0.9536	0.039	0.9594	0.020
supcon→res-nplm concat	0.9465	0.000	0.9574	0.000
supcon_sigreg→res concat	0.9135	0.011	0.9417	0.040
supcon→res residual alone	0.8587	0.026	0.8392	0.080

Table 7: Exp 73. CIFAR-100 supcon→res concat 0.9594 is a NEW C100 probe record (previous 0.9546, sup recipe); CIFAR-10 supcon→res concat ties the pre-discovery concat record (0.954). *This parent seed drew 0.9577 (above the exp-50 archive’s 0.9443 — a different seed path), so the honest residual effect is the +0.017 delta over its own parent. The objective split lands exactly where the transfer grid predicts: plain res > res-nplm on CIFAR (many-class, coarse/low-res — the DTD/galaxy10 side), and the plain-res residual half is the best-calibrated supervised C100 space on record (eucl 0.650, per-event 0.08). ss-parents again lose on both datasets.

0.321 → 0.530, mahaT 0.302 → 0.455. Probe records remain pre-discovery (C100 0.9594) or, for CIFAR-10, at exp-59’s high-purity post-discovery peak (0.984, split-dim sup→res-nplm).

5 Verdicts

1. **Substrate decides whether discovery helps.** On frozen trunks, fine-grained novelty is not geometrically outlying and discovery erodes the space; on task-fine-tuned trunks the same impure-pool fine-tune acts as refinement (50/72 cells positive; pool purity can stay low). supcon-ft+discovery is the only universally safe pipeline (12/12).
2. **Champion objectives are regime-determined and base-invariant.** Fine-grained many-class → SupCon+SIGReg; texture → SimCLR (labels add nothing: DTD record 0.854 is unsupervised); few-class coarse → SupCon for the probe and supervised distance-NPLM for calibration — the CIFAR class-count rule, verbatim, on transfer data.
3. **Residual training > discovery everywhere, and discovery on top of residuals is**

purity-gated. Residual records: cars 0.855 (+0.088 over discovery), flowers 0.885, galaxy10 0.975; a further discovery pass stacks only where the residual space separates the novelty, adding flowers $\rightarrow$ 0.906 and DTD $\rightarrow$ 0.862 (exp 72). The winning residual objective splits by regime (res-nplm on fine-grained, plain res on texture/coarse); supcon-ft is the universal parent, and ss-ft parents fail on fine-grained data because their SIGReg marginal has already absorbed the residual directions.

4. **The probe–calibration dissociation is real but not absolute:** NPLM spaces are the calibrated, distance-ready ones throughout, softmax spaces the probe-ready ones — and the res-nplm concat is the one construction found so far that delivers both at once.
5. **The recipe is world-independent** (exps 73–74): on CIFAR it reproduces every transfer-grid regularity — concat $>$ parent, supcon the right parent, the res/res-nplm objective split by regime, and discovery-on-residuals as purity-gated calibration repair — and sets the new CIFAR-100 probe record (0.9594).
6. **End-to-end fine-tuning erases the frozen-base ranking** (VISReg, weakest frozen, holds several pre records), extending the exp-43/49 inversion to the open-world battery.

Artifacts: scripts `experiments/70_cars_ft_suite.py`, `experiments/71_residual_ft_grid.py`, `experiments/72_residual_discovery.py`, `experiments/73_cifar_residual_ft.py`, `experiments/74_cifar_per-cell` logs and npz archives in `logs/exp70–exp74`; full tables in `docs/AIRCRAFT_MASTER_TABLE.md`; checkpoints `checkpoints/{ds}_ft_{base}_{arm}_seen.pt`.